Mixture-of-Prefetchers

A compact L2-cache router tuned by OpenEvolve

Trace behavior shifts
Best prefetcher changes by workload.

Route cheaply
Observe one 500K-instruction epoch, choose the next.

Tune policy
OpenEvolve edits compact routing constants.

Use one baseline
Prefetcher off is 1.000x in every plot.

What We Built On Athena

Contribution over Athena
Athena provides the simulator and L2C prefetchers. We add routing, epoch counters, fixed splits, OpenEvolve search, and rebuildable plots.

- Experts: MLOP + SPP+PPF at L2C.
- MoP-V1: one-epoch probe router.
- MoP-V2: OpenEvolve-tuned router.

Evaluation Protocol

- Prefetcher off is the 1.000x baseline.
- Best expert = max(MLOP, SPP+PPF).
- Split: 17 training, 7 heldout.
- Epoch: 500K retired instructions.
- IPC tracks cycles under fixed instructions.

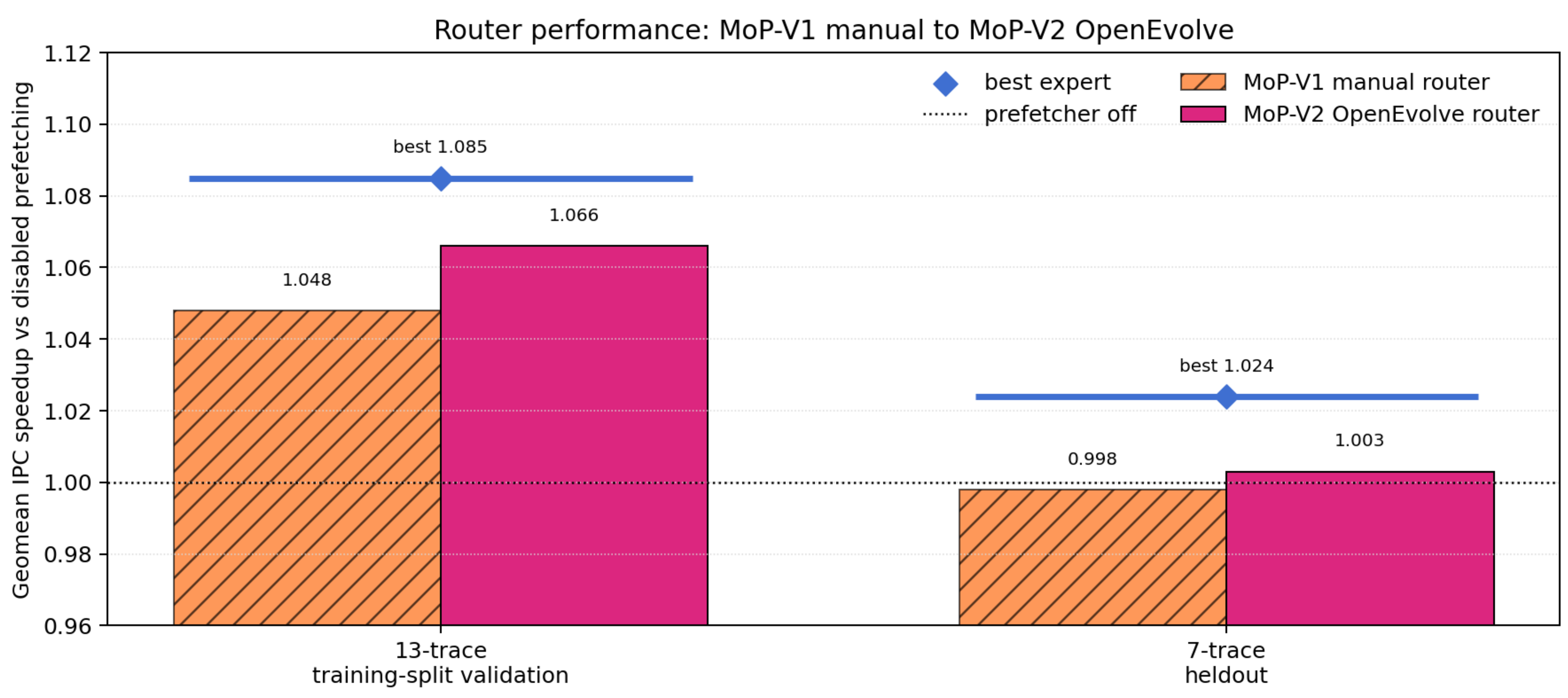
Two Athena Experts

MLOP	Learns useful address offsets across multiple lookahead distances. Better on 4 of 13 validation traces.
SPP+PPF	Tracks page signatures and delta paths, then filters candidates with a perceptron. Better on 9 of 13.

Trace Sets

Official	24 traces: SPEC, PARSEC, Ligra, secret_compute.
Training	17 for search. Final validation used 13 available traces.
Heldout	7 frozen traces after policy selection.
Window	Validation 500K + 1M. Heldout 20M + 50M.

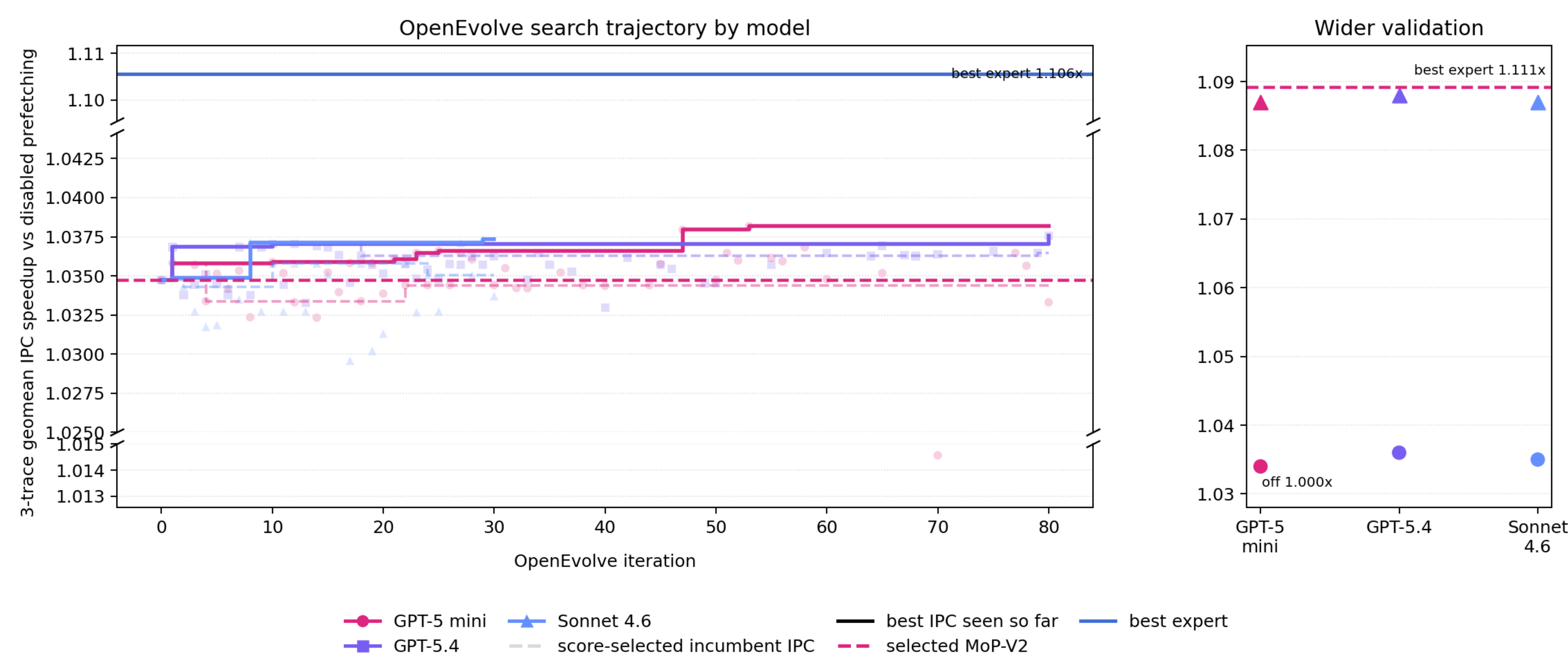
Before and after OpenEvolve



Baseline: disabled prefetching at 1.000x. Dark-blue diamond/line: per-trace best expert before geomean.

Caption: bars compare disabled prefetching, each single expert, the manual MoP-V1 router, the OpenEvolve-tuned MoP-V2 router, and the best expert reference on the same heldout and training-validation results.

Search trajectory by model

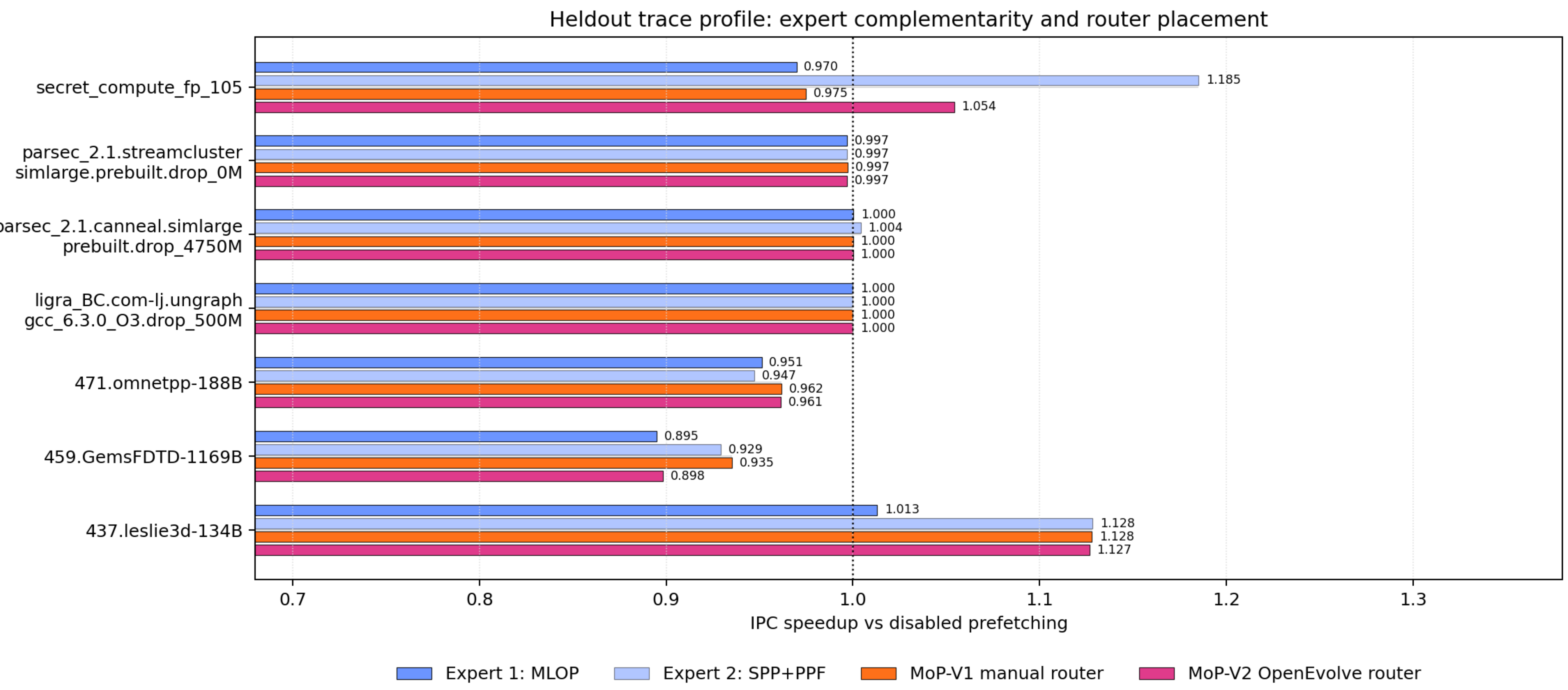


Faint points are candidates. Faint dashed lines show score-selected incumbent IPC. Solid lines show best IPC seen so far. Selected MoP-V2: 1.089x.

Caption: faint points are generated candidates. Solid traces show best IPC seen so far. Dashed lines show the score-selected incumbent IPC.

Heldout trace profile

Caption: each row is a heldout trace. Bars use disabled prefetching as 1.000x. MLOP and SPP+PPF sit above the MoP bars so the reader can see when the two experts disagree and where the router lands.



Each trace row is ordered top-to-bottom: Expert 1, Expert 2, MoP-V1, and MoP-V2. Black dotted line is disabled prefetching at 1.000x.

Source: final report Markdown. Figures rebuild from raw simulator summaries. Portrait poster generated by scripts/make_stage2_poster.py.